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Rumor Source Detection in Subgraphs: An ML Approach

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Abstract

Social networking sites offer a vast platform for sharing various types of content, such as information, photos, videos, and audio clips. However, the credibility of the information shared on these sites is a significant concern since there is no prior verification process as the information spreads across the networks. This increases the chances of disseminating false or misleading information, often called rumors. The spread of rumors in today's information environment creates severe problems for the credibility of information and public confidence. The quick spread of rumors, frequently conveyed via social media and other online platforms, can affect public opinion and potentially have real-world repercussions. Maintaining the integrity of information ecosystems and averting the potentially damaging impacts of disinformation require detecting and mitigating rumors. In today's social media and online communication age, it is crucial to identify rumors to prevent their spread and mitigate their potential negative impact. In this study, our primary goal is to discover and identify individual sources of rumors. This work addresses two aspects of rumor management: firstly, identifying the difference between rumors and facts, and secondly, tracking the source of the rumor. In conjunction with the attributes, state-of-the-art methods such as SVM, AdaBoost, Random Forest, Logistic Regression, and Graph Convolutional Networks (GCN) are used to accomplish these goals. This approach contributes to the broader goal of combating disinformation and preserving the Accuracy of information.

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Keywords: Centrality measures; Social network ; Machine Learning; Deep Learning; Rumor Source; Rumor Identification;

1. Introduction

The emergence of social networks has revolutionized the way we interact, communicate, and share information globally. Popular platforms such as Facebook, Twitter, Snapchat, and Instagram have made it easier for people to connect. Information spreads quickly in the age of social media, making it more challenging than ever to distinguish between reliable information and untrue rumors [1],[2]. The unrestrained spread of rumors and false information has emerged as a new and robust threat as a result of the communication revolution brought about by these networks. These untruths, which frequently come from a single source, can significantly affect how the public perceives things,

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how opinions are formed, and even how decisions are made [3]. Single-source rumors seriously affect social media, affecting specific users and society. These rumors can spread unjustified panic, stir up unneeded social or political unrest, and undermine public confidence in reliable information sources. As a result, it is now crucial to research this field and apply it to social media platforms, which calls for creative methods and a thorough understanding of the dynamics of online information.[4]. The spread of false information, the breakdown of trust, and the potential for social, political, and economic instability are just a few of the severe effects that rumors can have. Therefore, it is imperative to provide efficient techniques for identifying rumors on social networks, especially those with a single source.

Identifying rumors on social networks is challenging since it involves examining various factors, such as post content, user interactions, propagation patterns, and credibility evaluation [5]. Rumors might start from a single source, such as a bogus tweet or a made-up news piece, or they can spread from one source to many sources by being amplified by incorrect information. By concentrating on detecting rumors with a single source, false information can be located and contained before it spreads widely. Various research issues are associated with detecting single-source rumors in social networks. It is challenging to manually review and confirm each piece of information due to the abundance of user-generated content. As a result, automated rumor detection approaches utilizing machine learning, natural language processing, and network analysis techniques have become crucial tools.[6]

Rumors frequently have distinguishing traits that distinguish them from trustworthy information. Some features include emotive language, the lack of reliable sources, the quick spread across social networks, and the amplification by popular users. Finding rumors with a single source presents new difficulties. Contrary to rumors spread by several sources, single-source rumors necessitate an emphasis on locating the first source of untrue information. This entails studying distribution patterns and spotting distinguishing qualities or indicators that can be used to pinpoint the original rumor's creator.

This paper investigates methods for detecting the occurrence and origin of rumors in network subgraphs. These subgraphs contain the dissemination of information, which may or may not be rumors, and they serve as a time-limited picture of the nodes involved and their interactions. Single-source rumor identification in each of these subgraphs, using a variety of graph features, is done in this paper. Several features, including eccentricity, cycles, and centrality measurement, are examined to find the rumor-originating node. A brief discussion on the subsequent application—informing the nearby nodes of the rumor source—follows. The machine learning techniques, along with PCA and graph neural networks, are examined in predicting the presence of rumors in the subgraph. The subgraphs contain active nodes, which refer to the interacting nodes. This means that all the nodes in the neighborhood of the rumor node are not featured. The literature focuses on working on the interactions of active users in the subgraph on static graphs. The paper is organized as follows: Section 2 deals with related works, followed by rumor detection using centrality measures, ML models, and GCN in Section 3. Experimental analysis are discussed in Section 4, followed by a conclusion in Section 5.

2. Related Work

The current methods for identifying rumours can be categorised into three primary groups: propagation modeling-based, content-based, and context-based approaches.

2.1. Propagation Modelling Based Approach

The research based on observing the process of news proliferation over time is a *propagation-based approach*. The network epidemic models can explain how rumors spread by comparing them to the spread of infectious diseases. There is strong empirical support that false news spreads differently than actual news, creating patterns of distribution that might be used to detect false news automatically. In contrast to *content-based approach*, which is independent for each language, propagation-based features are likely to generalize across different languages, regions, and geographies. Furthermore, controlling how news spreads through a social network is typically outside the control of individual users, suggesting that propagation-based features could be complicated for hostile assaults to interfere. On Weibo, rumors are found using a GA-SA-ELLA model [7]. The three new propagation features and the BERT semantic feature were applied to improve the impact of rumor recognition. The latter is inferior to the best minimal feature

subset generated by GA and SA alone. However, as the model's proficiency increases, the GA-SA-ELLA model can learn tasks effectively even with less training data for each task. Nevertheless, this approach could not identify rumors at an early stage. The authors in [9] discuss detecting single rumors using the susceptible-exposed-infected-recovered (SEIR) paradigm. According to the SEIR paradigm, every node in the network is classified into four groups: susceptible (S), exposed (E), infected (I), and recovered (R). The Combined Network Centrality Approach (CNCA) [10] combines rumors' centrality with betweenness's centrality. The node with the highest CNCA value is the selected estimated source. Three other existing algorithms—rumor centrality, betweenness centrality, and Jordan centrality—are used to compare the results of the CNCA method.

2.2. Content Based Approach

Most studies on rumor detection primarily employ content-based methods that rely on linguistic traits, such as lexical and syntactical clues, to identify fraudulent writing styles. A significant limitation of content-based techniques is their susceptibility to sufficiently advanced fake news that may not be instantly recognizable as false. Moreover, the majority of linguistic characteristics are specific to a particular language, which restricts the use of these methods.

Textual rumor detection focuses on short, humorous, and suggestive texts. However, they are challenging for language processing systems to handle, necessitating the study of rumor propagation to cope with such posts. These letters may contain statements that seem sincere but frequently make indirect references. Such texts are challenging to represent in a language model and require high-level semantic analysis. As a result, attention is being given to how the rumor spreads on social media. A rumor detection system based on subject classification and content interpretation, focusing on the peculiarities of long text and the ease of fabricating rumor information in the health industry, is employed in [11]. Following subject categorization, the context logic of words and sentences within and between topics is taken into account, and training in synonyms and antonyms is done to improve the model's capacity to spot logical fallacies. It enhances the model's ability for generalization and the efficacy of rumor detection in the medical profession.

2.3. Social Context-Based Approach

User demographics, such as age, gender, political affiliation, educational status, and user reactions, such as posts that go along with news items or likes, are examples of social context features [12]. Social network structure refers to user connections, such as friendships or follower/followee relationships [14].

This paper utilized centrality metrics, ML models, and GCN as a reliable analytical framework for identifying and analyzing rumors. Using network theory, particularly centrality measurements, our objective was to pinpoint crucial nodes in the information transmission network. Through the evaluation of the importance of nodes using centrality metrics such as rumor centrality, closeness centrality, Jordan center degree centrality, betweenness centrality, and eigenvector centrality, our objective was to reveal the key elements in the network that have a critical role in the spread of rumors. By employing this methodology, we were able to identify influential sources and track the routes by which rumors disseminate, therefore augmenting the accuracy and efficacy of rumor identification in our inquiry.

3. Rumor Detection

3.1. Detection based on centrality measures

Both single-source and multiple-source rumors can be detected. In this study, rumor detection from a single source is considered. Since rumors propagate more quickly on connected components, the nodes in the largest weakly connected component are typically considered for rumor detection. The various centrality measures employed in this paper to detect rumors are

- Rumor Centrality
- Jordan Center
- Max Cycle Presence

- Closeness Centrality
- Estimation using Voronoi Cells
- Betweenness Centrality
- Eigen-vector Centrality
- Degree Centrality together with Closeness Centrality
- Eccentricity with Closeness Centrality

3.1.1. Rumor Centrality

The most popular method for identifying rumor sources is rumor centrality. Like most centrality measurements, the greater the rumor centrality, the more likely it is that a node is the rumor's source. A graph G is provided as input, and each vertex is given a score or positive integer. The source with the highest score or rumor centrality is then considered the estimated source. Since the BFS tree is being constructed at each node, this method doesn't need to choose the largest weakly connected component. The score is then determined using the BFS tree, which ensures that infinity or other undefined scores, which would be encountered using the other methods discussed, are avoided. The following procedures are used to determine a node's rumor centrality:

- Source node is chosen, Let it be u.
- Convert the graph into a BFS tree with root u using the Breadth First Search Technique.
- For the root node u, calculate the rumor centrality as per Equation 1.

$$R(u, G) = \frac{|V|!}{\prod_{w \epsilon V} T_w^u} \tag{1}$$

where $|V|$ is the size of the tree and T_w^u is the size of the Subtree of G that is rooted at w and points away from u. The node with the maximum rumor centrality will be the rumor source.

3.1.2. Jordan Center

The Jordan Center considers the nodes to be the greatest connected component. Because the Jordan center employs eccentricity as a distance measure, the nodes would have limitless distances if they existed outside its connected component. On social media, rumors frequently travel to nodes entirely outside of a locally connected community due to recommendations. Eccentricity, or the most significant distance between a node and any other node in the component, is computed for each vertex v in the largest linked component. In this strategy, the source node is assumed to have a minor eccentricity, and the Jordan center of the graph is calculated as 2.

$$J(G) = argmin_u \epsilon V(argmax_{w \epsilon V - \{u\}} dist(u, w)) \tag{2}$$

3.1.3. Max Cycle Presence

In this approach, an assumption is made that rumors frequently interact directly with the originating node and spread to several nodes. The most significant linked component was initially transformed into an undirected graph to apply this approach. This is because we are just interested in whether a cycle exists between two nodes and are not concerned with how the interactions occur. The related component's cycle basis is then discovered. An undirected graph's cycle basis is a collection of short cycles that is the foundation for the graph's cycle space. A graph's minimum spanning tree is used to determine the fundamental cycle foundation of the graph. One fundamental cycle, comprised of the edge, its end nodes, and the path in the minimal spanning tree that connects them, exists for each edge, not the minimum spanning tree. They form a compact representation of the set of all cycles when taken as a whole. Next, the source node is identified by looking at the node participating in most cycles. This is predicated on the fundamental presumption that nodes primarily communicate with our source node in cycles. If there are only a few cycles, say one

cycle on a cycle basis, the decision has to be made about which node to employ; proximity centrality as a tie-breaker is used here.

3.1.4. Closeness Centrality

This approach is entirely dependent on a graph characteristic. The closeness centrality for each node in the subgraph is determined, and the node with the highest closeness centrality is chosen as the source. The most significant component that is weakly connected is chosen. The average distance between a node and other nodes will probably become infinite if the complete subgraph is considered because there are disconnected components. An approach to identify nodes that can efficiently spread information throughout a graph is through their closeness centrality. Since closeness centrality is the inverse of average distance, the closer a node is to other nodes in the network, the higher its closeness centrality. Since most other nodes communicate with the source node in the largest weakly linked component and then propagate the rumor to their neighbors, we anticipate a source node to have this capability. The Equation 3 is used to determine a node's closeness centrality.

$$C(x) = \frac{N-1}{\sum_{y \epsilon V} d(x,y)} \tag{3}$$

N is the number of nodes, and d(x, y) is the distance or path length between the two nodes. The scores are normalized. The subtraction with 1 in the numerator can be avoided for large networks, but here, it is retained since it is a smaller network.

3.1.5. Estimation using Voronoi Cells

This technique is an adaptation of [13],[17], which was used to identify many sources in a network. Modifications are done to identify a single source. With this approach, we aimed to enhance the closeness centrality method's predictions. The two nodes with the highest closeness centrality are first identified by picking the largest weakly connected component. Therefore, these nodes are considered the central nodes for the Voronoi partition. Based on the shortest-path distance metric, the Voronoi cell centered at a node c is the collection of all nodes v that are closer to c than any other center node in set C. Here, C represents a set of nodes in the graph, and c is one of the elements in C. The source node is estimated by identifying the center node with the largest Voronoi cell.

3.1.6. Betweenness Centrality

The so-called betweenness centrality measure determines a vertex's position relative to other vertices on the shortest pathways between them. Stated differently, it facilitates the identification of individuals operating within a network in "bridge-spanning" roles. We try to determine whether substantial information may be used to identify the source node flowing across it in contrast to nodes elsewhere. The equation 4 is used to determine a node's betweenness centrality.

$$B(v) = \sum_{s \neq v \neq t} \frac{\delta_s t(v)}{\delta_s t} \tag{4}$$

where $\delta_s t$ is the total number of shortest paths from s to t, and $\delta_s t(v)$ is the total number of shortest paths passing through v.

3.1.7. Eigen vector Centrality

Eigenvector centrality is a method for measuring the transitive impact of nodes. It represents how substantial an entity is based on how significant the entities it interacts with are. The eigenvector centrality, in other words, reveals which nodes in the network have stronger ties to other important nodes. Connections from high-scoring nodes boost a node's score more than connections from nodes with lower scores. Numerous other nodes with high scores are

connected to a node with a high eigenvector score. The node with the highest eigenvector centrality is, therefore, the most influential and is therefore taken to be the source among all the nodes with high scores. Eigenvector centrality is used to identify the information source in the subgraphs. Like the earlier method, the source node is considered the node with the highest eigenvector centrality. This helps us to assess if the source node is the most potent node based on this statistic and if other vital nodes surround it. Therefore, in our subgraphs, a source node is thought to have the most influence.

3.1.8. Degree Centrality plus Closeness Centrality

A straightforward centrality metric called the degree counts each node's neighbors. There are two variations of the metric if the network is directed: in-degree, which is the number of incoming links, and out-degree, which is the number of outgoing links. In-degree is considered since other network nodes provide in-links, and the node determines out-links. This also aligns with our hypothesis that many infected nodes communicate with the originating node after hearing a rumor. In this technique, the nodes with the highest in-degree centrality are chosen. These nodes are compared for their closeness centralities, and the node with the highest closeness centrality is then determined to be the source node. Therefore, the nodes with high proximity centrality, many interactions, and the ability to quickly spread rumors around the network are identified.

3.1.9. Eccentricity plus Closeness Centrality

The maximum network distance between a graph vertex's v and any other vertex's u in a connected graph component G is known as the vertex's eccentricity. All vertices in a disconnected network are assumed to have infinite eccentricity. The nodes with the lowest eccentricities are chosen, and their maximum closeness centrality is examined.

3.2. Rumor Detection using Machine Learning Models

The presence of rumor and the source itself could not be determined explicitly using the method outlined, as was stated in the previous section. Our algorithms accurately estimated sources for each of the three categories. As a result, the subgraphs are divided into those that contain rumors and those that do not. For those subgraphs that have been determined to include rumors, the issue can be modeled as a binary classification problem followed by source discovery [14]. Graph classification is applied to them to identify the presence of rumors in the subgraphs. The following two strategies are employed:

3.2.1. Using graph attributes to train machine learning classification models.

Data pre-processing and data extraction: The data is presented as graphs. All of the graphs were transformed into a tabular format. This format includes the attributes of the graph as well as a class label of 1 for graphs that contain rumors and 0 for those that do not. The graphs are used to extract 11 attributes. The attributes are:

- Nodes: The count of users in the subgraph.
- Edges: This corresponds to the number of interactions in the subgraph.
- Degree Centrality: The average of all degree centrality of the graph's nodes. It is the nodes' normalized degree.
- Closeness Centrality: Closeness Centrality value's average across all the subgraph nodes is calculated.
- Eigenvector Centrality: The average of eigenvector centrality over all the nodes is taken for each subgraph.
- Core number: The sub-graph's average number of core nodes is calculated. The vertex's core number is the greatest k value in a k-core containing a vertex.
- Average Clustering Coefficient: The average clustering coefficient of the subgraph.
- Transitivity: The average value of transitivity of nodes in the subgraph.
- Square Clustering Coefficient: The mean of this value of all the nodes.
- Radius, Diameter: Included the average radius and diameter of the nodes in the subgraph to train our ML models.

3.2.2. Data Preprocessing

- One vs. One Classification: In this study, the One-vs-One (OvO) classification is investigated for the problem of rumor classification. The data set consists of three categories: two are related to rumors, and the third is related to

non-rumors. Training and testing are done in two scenarios to guarantee strong model performance, preventing a model from overfitting a particular data set. The study finds the best models to solve the "5G-Conspiracy vs. Non-conspiracy" classification problem, highlighting the top-performing models. A comparative study is conducted for the other two classification challenges, "Other-Conspiracy vs. Non-Conspiracy," to ascertain which model performs better in each scenario. Lastly, the study concludes with the best-performing models.

- Hyper-parameter Tuning using Grid Search: Hyper-parameter tuning utilizing grid search was done to find the ideal hyper-parameter settings for machine learning models to maximize the models' performance. To fine-tune the models, grid search was used to methodically investigate a preset set of hyper-parameter values, such as learning rates, regularisation strengths, and feature selection criteria. Through a thorough evaluation of every possible combination of hyper-parameters and measurement of their effect on performance metrics, like Accuracy and precision, the study ensured the models were well-suited to the unique features of tip data. This method improved the model's ability to distinguish between genuine and false rumors, leading to more reliable and accurate detection of rumors.
- Cross-Validation: When working with data sets with imbalanced class distributions—where some classes are underrepresented compared to others—stratified k-fold cross-validation is very useful. This approach helps guarantee that the model is trained and assessed on a balanced subset of the data by preserving class proportions in each fold. This is particularly important for applications where the minority class is crucial and needs a model to identify it, such as fraud detection, medical diagnostics, or unusual event prediction. By increasing the model's capacity to generalize to actual situations, stratified k-fold cross-validation lowers the likelihood of overfitting to the majority class and offers a more accurate evaluation of the model's performance. It is a flexible method that enhances the machine's resilience and dependability. Our methodology uses cross-validation, more precisely, a five-fold stratified cross-validation strategy, for model training. Using k-fold cross-validation, the training dataset is divided into k smaller, equally stratified subsets. Although there are other approaches (explained below), they all work from the same fundamental ideas. The following actions are performed for every one of the k "folds" or subsets:
  - A model is trained using the k-1 folds as training data.
  - The resulting model is validated on the remaining 1 part of the data.

  The average of the numbers calculated in the loop is then the performance indicator supplied by k-fold cross-validation. Though computationally intensive, this strategy does not generate a lot of data waste. The stratified K-fold was utilized because there is a class imbalance. This K-Fold variant yields stratified folds. The folds are created by keeping track of the sample percentages for each class. As a result, no splits solely include the dominant types.
- Dimensionality Reduction: To minimize the data dimension, principal component analysis is employed. As much variance from the initial data is preserved using this strategy, over 90 percent of the information is captured for all three kinds of information. Therefore, the dimensions are decreased to 7. The models that were used for rumor detection are Support Vector Machine (SVM), Random Forest, AdaBoost, and Logistic Regression.

3.2.3. Rumor detection using GCN.

This model categorizes all the graphs into rumors and non-rumors using Graph Convolution Networks. All the nodes in the subgraphs are the same because they are homogeneous. Two features from the dataset are present in the nodes. The number of friends and followers the user connected to the node has is represented by the friends and followers count. This is not the nodes' in-degree and out-degree. For each user across the entire data set, they are constant.

The Stellar-Graph library, a toolkit based on TensorFlow and Keras, is used to create and train our model. A well-liked option for deep learning applications, the Adam optimizer trains the model. The benefits of the Adagrad and RMSProp optimizers are combined in the Adam optimizer. A 0.001 learning rate is applied. During optimization, the step size is regulated by the learning rate. Selecting the correct learning rate is crucial to ensuring the model converges efficiently. Hyper-parameter tweaking determined the best learning rate for the particular task. 32 is the batch size used to train the model. Mini-batch Gradient Descent is used to reduce convergence time and maximize memory utilization. The particular dataset and the available computer power determine the batch size. A task for classifying graphs predicts an attribute for each graph in the collection. For instance, they categorize each graph (binary or multi-

Method	Accuracy on 5G Class in %	Accuracy on Other Class in %	Accuracy on Non-Conspiracy Class in %
Rumor Centrality	27	21	18
Jordan Center	34	34	50
Max Cycle Presence	41	57	51
Closeness Centrality	72	79	81
Estimation Using Voronoi Cells	56	52	0
Betweenness Centrality	49	48	37
Eigenvector Centrality	60	63	56
Degree + Closeness	72	80	81
Eccentricity + Closeness	7	3	3

Table 1: Accuracy of centrality measures in estimating the source of subgraphs.

class classification) or make a regression prediction of a continuous value. It is supervised, meaning a subset of graphs with ground-truth labels are used to train the model. The graph's adjacency and node feature matrices serve as the input. Each of the first two layers of the graph convolutional algorithm has 64 units and employs *ReLu activation*. The learned node representation is compiled to generate a graph representation in the following layer: a mean pooling layer. Two completely connected layers with 32 and 16 units and *R*eLu activation, respectively, are fed the graph representation. The final layer is the output layer with a single unit and *S*igmoid activation.

4. Experimental Analysis

A labeled dataset, the Wico Graph Dataset [16], is used for the experiment. The dataset consists of three classes.

- The 5G-Corona Conspiracy: This class encompasses tweets that assert or imply a significant link between COVID-19 and 5G technology. These tweets may suggest that 5G weakens the immune system, leading to the current pandemic, or propose that the pandemic is non-existent and that COVID-19 victims were harmed by radiation emitted by 5G network towers. The essential prerequisite is the asserted presence of a causal connection.
- Other Conspiracy: This category encompasses all tweets that propagate conspiracy ideas not previously mentioned. This encompasses notions of deliberate dissemination of the virus, compulsory or detrimental immunizations, or the virus being a fabrication.
- Non-Conspiracy: This class includes all tweets that do not fall within the categories of the preceding two classes. It is essential to acknowledge that this encompasses tweets that specifically address the COVID-19 pandemic, tweets that assert that the safety of 5G technology has not been definitively established or may even pose harm without directly connecting it to COVID-19, and tweets that allege that authorities are promoting the implementation of 5G. At the same time, COVID-19 diverts the public's attention.

Each node is given an ID distinct across all subgraphs and displays the node's or user's number of friends and followers. In the Graph, friends and followers will be used. Four hundred thirteen subgraphs are in the first category, 598 in the second, and 2502 subgraphs in the third category.

Models	5G vs Non-Conspiracy		Other vs Non-Conspiracy		Non-Conspiracy vs Conspiracy	
	Precision	Recall	Precision	Recall	Precision	Recall
Without PCA						
SVM	0.46	0.15	0.30	0.22	-	-
Logistic	0.21	0.01	0.26	0.01	-	-
Random Forest	0.34	0.13	0.34	0.17	-	-
AdaBoost	0.43	0.07	0.51	0.11	-	-
With PCA						
SVM	**0.60**	**0.12**	**0.60**	**0.07**	**0.62**	**0.20**
Logistic	0.36	0.01	0.20	0	-	-
Random Forest	0.37	0.15	0.36	0.18	-	-
AdaBoost	0.36	0.01	0.20	0	-	-

Table 2: Results of Machine Learning Classification Task

4.1. Inferences obtained from Centrality Measures

Table 2 shows the Accuracy of each method employed. Compared to all other techniques, closeness centrality is the best indicator of the source in a tiny subgraph of interactions. It does not appear that eccentricity is a meaningful metric. However, it showed poor results when applied to the Jordan Center approach. Additionally, it utterly destroyed proximity centrality's efficiency, which was otherwise doing an excellent job on its own. A good result was produced through Eigenvector Centrality. This can suggest the presence of essential spreader nodes in the rumor's spread near the source nodes. As a result, nodes in direct contact with the potential rumor source and their neighbors should be alerted. The efficiency of closeness centrality was not significantly increased by degree centrality alone. The nodes that have the potential to spread rumors also appear to have a higher degree in the interaction subgraph that was taken as a snapshot across time since degree centrality was used to choose the nodes to be tested for closeness centrality. Compared to closeness centrality, the strategy based on source nodes in numerous cycle bases did not produce satisfactory results. Even though this probably shows how ineffective this approach is, this alone might not be able to determine whether source nodes are included in many cycles. This is because the actual number of cycles may be significantly higher than the number of cycles calculated on a cycle basis. Betweenness centrality is a poor indicator of information sources. It shows that a node's capacity to regulate the information flow across it does not imply that it is the source. Interestingly, employing Voronoi partitions to search for information seemed to work somewhat on the rumor subgraphs but yielded an accuracy of 0 for non-rumor subgraphs. Furthermore, the Accuracy of rumor centrality was relatively poor. This can mean that the measure is inappropriate for graphs with small scales. Table 2 shows the results of each of the machine learning models.

4.2. Analysis on the Machine Learning and GCN Models

SVM consistently outperformed Random Forest, Ada Boost, and Logistic Regression in all three scenarios. On the final data set, it was 0.62% precise. We observed that 0.62% of the classes are observed as rumor classes. Although an algorithm used in a real-world setting would need to be more accurate, it can be made better with a larger dataset. The algorithms' recall scores were, however, low. This indicates that it could only recognize a limited number of true rumors. A graph neural network architecture mimics identical circumstances since the classification results of machine learning models are not very promising. In repeated k-fold cross-validation with k = 10, GCN provided 71.3% accuracy. The GCN model could handle a larger dataset. A larger dataset will be needed to fine-tune it and reduce any overfitting that might have occurred.

5. Conclusion

In this work, we used a dataset of three classes- two corresponding to rumors and one to non-rumors—to detect rumors. We investigate how different classification algorithms like Support Vector Machines (SVM), Ada Boost, Ran-

dom Forest, Logistic Regression, and Graph Convolutional Networks (GCN) handled the multi-class task. Based on our investigation, SVM performed the rumor detection task better. GCN, a neural network-based method explicitly designed for graph data, demonstrated how it might tackle rumor detection differently. This implies that investigating graph-based ways is crucial when handling the propagation of rumors in social networks or other linked data. The present study underscores the possibility of further investigation to enhance rumor detection techniques. Hybrid techniques that incorporate the best features of several algorithms could be explored further and lead to improved classification performance.

References

[1] Nair, Lekshmi S., and Jo Cheriyan. "SIR-M Epidemic Model: A SARS-CoV-2 Perspective." Computational Intelligence and Data Analytics: Proceedings of ICCIDA 2022. Singapore: Springer Nature Singapore, 2022. 499-508.

[2] Shelke, Sushila, and Vahida Attar, "Rumor detection in social network based on user, content and lexical features." Multimedia Tools and Applications 81.12 (2022): 17347-17368.

[3] Sun, Shengyun, et al, "Detecting event rumors on sina weibo automatically." Asia-Pacific web conference. Springer, Berlin,Heidelberg, 2013.

[4] Yu, Suisheng, Mingcai Li, and Fengming, "Rumor identification with maximum entropy in micronet." Complexity 2017 (2017).

[5] Suthanthira Devi, P., and S. Karthika. "Rumor Identification and Verification for Text in Social Media Content." The Computer Journal 65.2 (2022): 436-455.

[6] Shelke, Sushila, and Vahida Attar. "Source detection of Rumor in a social network–a review." Online Social Networks and Media 9 (2019): 30-42.

[7] Yousheng Zhou, Chujun Wu, Qingyi Zhu, Yong Xiang, Seng W Loke, "Rumor Source Detection in Networks Based on the SEIR Model", IEEE Access.

[8] Dhanya, N. M., and U. C. Harish. "Detection of Rumors in Tweets Using Machine Learning Techniques." International Conference on Automation, Signal Processing, Instrumentation and Control. Singapore: Springer Nature Singapore, 2020.

[9] Abhijit Das, Anupam Biswas, "Rumor Source Identification on Social Networks: A Combined Network Centrality Approach", Advances in Intelligent Systems and Computing.

[10] Jing Gao, Xuan Sun, Li Tan, Zihao Ma, "Rumor Detection Based on Generalized Text Feature", Proceedings of the 2021 5th International Conference on Electronic Information Technology and Computer Engineering.

[11] Zhenyu He, Ce Li, Fan Zhou, Yi Yang, "Rumor Detection on Social Media with Event Augmentations", Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval.

[12] Yuan Gao, Xiang Wang, Xiangnan He, Huamin Feng, Yongdong Zhang, "Rumor Detection with Self-supervised Learning on Texts and Social Graph", Social and Information Networks.

[13] Santhosh, Nikita Mariam, Jo Cheriyan, and Lekshmi S. Nair. "A Multi-Model Intelligent Approach for Rumor Detection in Social Networks." 2022 International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS). IEEE, 2022.

[14] Athira Nair, K., Harikumar, G., Vissutha, M.P., Ajanalakshmi, D., Deepthi, L.R. "Classification of Trust in Social networks using Machine Learning algorithms", Proceedings of the 2022 3rd International Conference on Intelligent Computing, Instrumentation and Control Technologies: Computational Intelligence for Smart Systems, ICICICT 2022.

[15] Deepthi, L. R., et al. "A Novel Approach for Prediction of Air Pollutant Concentration." 2020 4th International Conference on Trends in Electronics and Informatics (ICOEI)(48184). IEEE, 2020.

[16] Pogorelov, Konstantin, et al. "Wico text: a labeled dataset of conspiracy theory and 5g-corona misinformation tweets." Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks. 2021.

[17] Nair, Jayashree, S. S. Akhil, and V. Harisankar. "Fake News Detection Model for Regional Language." 2022 IEEE 7th International conference for Convergence in Technology (I2CT). IEEE, 2022.

[18] MR, Neethu, and N. Harini. "Secure request response (SRR): a framework to classify trust/distrust relationships in social networking." Computing in Engineering and Technology: Proceedings of ICCET 2019. Springer Singapore, 2020.